

# RELIANCE

## Reliable and Certifiable Human-Centred Agentic AI for Structural Systems

New structural design, existing-structure assessment and retrofit, and lifecycle management of load-bearing systems

RELIANCE is a COST Action proposal on the responsible use of AI agents across two complementary structural-engineering missions: the **design of new structures** and the **identification, assessment and retrofit of existing structures**. Agents will coordinate trusted engineering tools, data and, where relevant, physical sensing or robotic systems. Their reliability and safety will be evaluated against established engineering practice, while the engineer retains authority over assumptions, verification and final decisions. Education and training are integral to the Action.

**Call**

COST Open Call OC-2026-1

**Collection date**

28 October 2026, 12:00 CET

**Action duration**

Four years

**COST support**

Networking, coordination, mobility and training

**Proposer network**

At least 7 COST Full/Cooperating Members; at least 50% of represented countries from ITCs

**2026 rule**

At least 40% of proposers must be YRIs

### Proposal contacts

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New structural design · existing-structure assessment · structural identification · monitoring · retrofit · lifecycle management

## 1. Context, challenge and scope

Structural engineers are beginning to work with AI systems that can do more than generate text: they can plan a sequence of actions, call analysis and code-checking software, combine information from different sources and inspect intermediate results. This creates opportunities at both ends of the structural lifecycle. For **new structures**, agents can support the progression from structural concept to analysis, option comparison, code checks and design documentation. For **existing structures**, they can combine inspection, monitoring, structural identification and model updating to support assessment, retrofit and lifecycle decisions. In both cases, the same engineering question arises: *what evidence is required before an agent-supported recommendation can be relied upon?*

For structural use, technical plausibility alone is insufficient. The engineer needs to understand the assumptions used, the tools called, the origin of the data, the uncertainty in the result, the checks that have been performed and the circumstances in which the AI output should be rejected or escalated for review. RELIANCE will therefore treat reliability as a property of the **whole decision process**: the agent, the engineering tools it invokes, the available data, and the human who reviews and accepts the result.

**The gap addressed by RELIANCE.** Europe does not yet have a shared structural-engineering framework for evaluating agentic AI in safety-critical decision chains. Work is advancing in AI, reliability, computational mechanics, monitoring, BIM, digital twins, robotics and human factors, but the evidence is fragmented and difficult to compare across tools, codes and structural applications.

### Why a COST Action

The main need is coordination. Relevant research already exists in universities, research centres, companies and public organisations, often supported through separate national or European projects. A COST Action can connect this work without duplicating it: the network can agree terminology, define benchmark problems, compare verification approaches, organise mobility and Training Schools, and bring practitioners, infrastructure owners and standards communities into the discussion from the outset.

### Two complementary engineering missions

#### Design of new structures

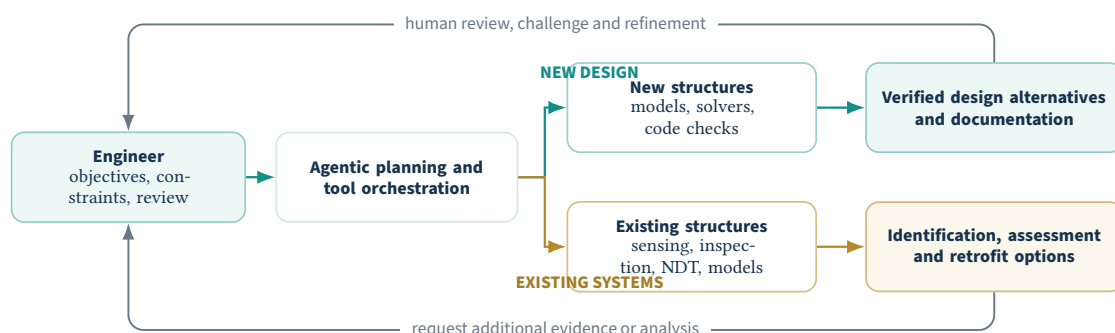
Agents can support structural-system exploration, load-path reasoning, analysis-model generation, solver and code-checking calls, option comparison, reliability-aware screening and transparent design documentation. The objective is not automated design for its own sake, but a controlled environment in which engineering alternatives can be generated, checked and reviewed efficiently.

#### Existing structures: assessment and retrofit

Existing buildings and infrastructure start from incomplete knowledge. Agents can combine records, inspection, monitoring, structural identification and model updating; identify missing information; plan further measurements; estimate condition and residual capacity; and compare repair, strengthening or retrofit options under explicit uncertainty.

### Digital and cyber-physical agentic loops

For new structures, the loop is primarily digital: agents can orchestrate structural models, solvers and code-checking tools, compare alternatives and return a transparent evidence trail to the engineer. For existing structures, the same logic extends to the physical asset through sensor networks, UAVs, mobile robots and non-destructive testing devices. An agent can request additional information, adapt an inspection plan or update a structural model as evidence accumulates. In both cases, the engineer defines the objective, reviews the evidence and retains authority over the final structural decision.



## Reliability, safety and the human role

The Action will not assume that an AI-assisted workflow is safer simply because it is automated. Reliability has to be demonstrated on common engineering tasks and compared with appropriate baselines. RELIANCE will therefore examine **human-only**, **agent-supported** and, where scientifically useful, **agent-only** performance on shared benchmark cases. The aim is to understand where agents reduce error or improve consistency, where they introduce new failure modes, and where human judgement remains essential.

### Compare performance

Common cases will allow comparison of error detection, decision quality, consistency and efficiency across conventional engineering and AI-supported workflows.

### Measure safety-relevant behaviour

Evaluation will include robustness to incomplete or noisy information, uncertainty calibration, missed and false warnings, repeatability, failure detection and the consequences of incorrect recommendations.

### Define human authority

Engineers set objectives and constraints, review assumptions and evidence, can challenge or override agent recommendations, and remain responsible for final acceptance. Escalation rules will define when additional analysis, data or independent review is required.

The central question is therefore not whether an agent can replace a structural engineer, but *under what conditions a human–AI team can perform more reliably than a conventional workflow without weakening professional accountability*. This includes trust calibration: uncertainty and verification status must be visible enough for the engineer to know when confidence is justified and when it is not.

## Scope

|                                | Within scope                                                                                                                                                                                                                                        | Outside the core scope                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>New structures</b>          | Structural-system exploration; conceptual and preliminary design; analysis and model generation; code checks; option comparison; reliability-aware screening; design documentation; selected constructability and design-to-as-built links.         | Architectural generation without a structural-safety objective; generic optimisation detached from engineering verification. |
| <b>Existing structures</b>     | Inspection and monitoring; structural identification and model updating; condition and safety assessment; residual-capacity evaluation; repair and retrofit; structural digital twins; risk-informed lifecycle decisions.                           | Generic building operation, HVAC, indoor comfort and non-structural services.                                                |
| <b>Cross-cutting assurance</b> | Verification and validation; reliability and uncertainty; comparison with human and conventional engineering baselines; trust calibration; human review and override; audit trails; code interaction; documentation; pre-normative recommendations. | Generic AI adoption without a structural-safety question; development of a single proprietary AI product.                    |
| <b>Construction interface</b>  | Structural quality control, as-built verification and selected human-robot interfaces where they test the same agentic assurance questions.                                                                                                         | Autonomous construction as a primary research objective.                                                                     |

## Relationship to adjacent COST Actions

RELIANCE complements existing COST communities while maintaining a distinct structural-engineering focus. **CA25158 Tech-TO.BE** addresses digital transformation across the wider AECO value chain, while **CA25178 API-Net** focuses on embodied and physical agentic AI and robotics. In RELIANCE, physical AI enters where agents task or coordinate sensing and robotic systems around a structure. RELIANCE focuses on **agentic support for the design of new structures and for the identification, assessment and retrofit of existing structures**, under a common framework for reliability, verification and professional decision-making.

## 2. Ambition, objectives and Working Groups

**Ambition.** RELIANCE will establish a European reference framework for human-centred, reliability-aware and certification-oriented agentic AI across the structural lifecycle. It will cover both **new structural design** and **existing-structure identification, assessment and retrofit**, defining how reliability and safety are demonstrated, how agent-supported performance is compared with established engineering practice, and how human judgement, traceability, validation and documentation are preserved for professional review and future standardisation or certification pathways.

## Research Coordination Objectives

|             |                                                     |                                                                                                                                                                                                                                                                   |
|-------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RCO1</b> | <b>Common language and reference architecture</b>   | Define terminology for agent roles, tool use, provenance, human checkpoints, decision authority and failure states.                                                                                                                                               |
| <b>RCO2</b> | <b>Benchmarking and comparable evidence</b>         | Agree benchmark problems for new structural design and for existing-structure identification, assessment and retrofit, together with validation hierarchies, failure modes and reproducibility requirements.                                                      |
| <b>RCO3</b> | <b>Reliability, safety and human-AI performance</b> | Coordinate methods for robustness, uncertainty and risk thresholds; compare conventional, agent-supported and human-AI team performance on common tasks; define deterministic cross-checks, trust-calibration measures, override conditions and escalation rules. |
| <b>RCO4</b> | <b>Digital and cyber-physical workflows</b>         | Define trusted interactions between agents and analysis/code-checking tools for new design, and between agents, digital twins, monitoring systems, UAVs, robots and non-destructive testing devices for existing systems.                                         |
| <b>RCO5</b> | <b>Assurance and pre-normative pathways</b>         | Develop documentation patterns and recommendations relevant to professional review, procurement, standards and codes of practice.                                                                                                                                 |

## Capacity-Building Objectives

**CBO1 - Build critical mass.** Connect structural designers, computational mechanics and reliability specialists, experts in existing structures and retrofit, system-identification and monitoring communities, AI researchers, robotics experts, human-factors researchers, software developers, practitioners and standards stakeholders.

**CBO2 - Educate the next generation.** Establish recurring Training Schools, mobility, mentoring and leadership opportunities for YRIs, with strong participation from ITCs.

**CBO3 - Strengthen exchange with practice.** Involve infrastructure owners, consultancies, SMEs, software providers and public authorities in benchmark definition and guidance.

**CBO4 - Leave reusable knowledge.** Produce open terminology, benchmark definitions, teaching material, case-study templates and professional guidance that remain useful beyond the Action.

## Working Group structure

### **WG1 - Trust, reliability, verification and human oversight**

System and human-AI team reliability; uncertainty; verification and validation; human baselines; failure modes; trust calibration; review, override and auditability.

### **WG2 - Structural design, assessment and retrofit**

New-structure design workflows; existing-system identification and model updating; code checks; safety and residual-capacity assessment; retrofit options; engineer review.

### **WG3 - Autonomous inspection, monitoring and cyber-physical interaction**

Agent-directed sensing and inspection, SHM, digital twins, UAV/robot/NDT interaction, lifecycle decisions and selected as-built or construction interfaces.

### **WG4 - Standards, education and uptake**

Pre-normative recommendations, stakeholder engagement, Training Schools, YRI development and open educational resources.

A small set of shared benchmark problems will connect the WGs. Two anchor benchmarks will represent the principal engineering missions: **(i) design of a new structural system**, from concept and analysis to code checking and documented option selection; and **(ii) assessment and retrofit of an existing structure**, from uncertain information and structural identification to safety evaluation and intervention choice. Wherever feasible, the same cases will be used to compare conventional human workflows with agent-supported workflows, so that improvements in consistency, error detection, decision quality and effort can be assessed alongside new failure modes. Additional benchmarks will address autonomous inspection or monitoring leading to an asset decision, and design-to-as-built verification. Construction remains a supporting interface, focused on structural quality control, as-built verification and selected human-robot tasks.

### 3. Programme, outputs and education

The four-year programme is organised around a limited set of shared outputs. Working Group meetings will support technical convergence, STSMs and Virtual Mobility will enable focused exchanges and reproducibility exercises, and Training Schools will turn the emerging methods into transferable skills.

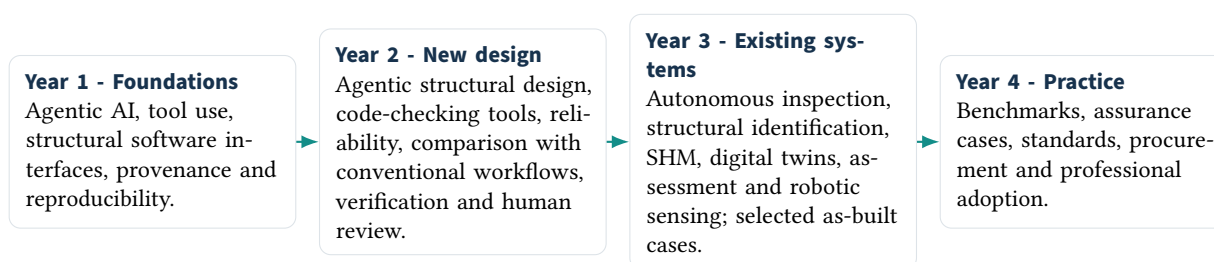
#### Outputs and timing

| ID        | Target | Output                                                                                                                                                                                                              |
|-----------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>O1</b> | Year 1 | European terminology and reference architecture for agentic AI in structural engineering, covering both new design and existing-structure assessment.                                                               |
| <b>O2</b> | Y1–Y2  | Benchmark specifications for new structural design and for existing-structure identification, inspection, assessment and retrofit, with a common evaluation matrix and representative failure modes.                |
| <b>O3</b> | Year 2 | Reliability and safety evaluation protocol covering verification, validation, reproducibility, uncertainty, risk control and comparison with conventional engineering baselines.                                    |
| <b>O4</b> | Y2–Y3  | Human-in-the-loop framework defining decision authority, trust calibration, review and override, escalation, provenance and assurance documentation.                                                                |
| <b>O5</b> | Year 3 | Reference workflow patterns for agents interacting with trusted analysis and code-checking tools in new design, and with digital twins and physical sensing or inspection systems in existing-structure assessment. |
| <b>O6</b> | Y3–Y4  | Pre-normative roadmap for standards, codes, procurement and professional acceptance.                                                                                                                                |
| <b>O7</b> | Annual | Training School material and an open RELIANCE curriculum for doctoral, university and professional education.                                                                                                       |
| <b>O8</b> | Year 4 | Synthesis white paper and implementation roadmap for reliable agentic AI in new structural design and existing-structure assessment, retrofit and lifecycle management.                                             |

#### Education mission

The educational programme is part of the technical work. Engineers and researchers need to learn how to use agentic systems across both major application domains: creating and checking design alternatives for new structures, and combining data, models and autonomous sensing for existing-structure assessment. Across both, the emphasis is on verification, uncertainty, traceability and professional responsibility, including how to compare AI-supported results with conventional engineering judgement, how to recognise over-confidence or weak evidence, and when to challenge, override or escalate an agent recommendation.

RELIANCE will organise a recurring Training School, with the theme evolving as the Action matures:



Each School will combine lectures with practical exercises and will generate material that can be reused in doctoral programmes, university courses and continuing professional education. YRIs will be encouraged to progress from participants to exercise developers, tutors and later programme leaders.

## How the networking tools will be used

|                                   |                                                                                                                                          |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WG meetings and work-shops</b> | Agree terminology, benchmarks, protocols and cross-WG decisions.                                                                         |
| <b>STSMs and Virtual Mobility</b> | Reproduce workflows, compare tools, develop shared methods and transfer know-how between institutions.                                   |
| <b>Training Schools</b>           | Build technical capability and test reusable teaching material.                                                                          |
| <b>Stakeholder work-shops</b>     | Test benchmark relevance, evidence requirements and uptake barriers with practice, owners, public authorities and standards communities. |
| <b>Communication activities</b>   | Share validated outputs with research, professional, education and policy audiences.                                                     |

## 4. Building the proposer network

RELIANCE will be built around partners taking responsibility for defined scientific, educational and stakeholder contributions. Relevant contributions include new-design and existing-structure benchmark problems, methods, data and tools, stakeholder connections, Training School modules and coordination roles.

### Partner profiles

| Profile                                                   | Possible contribution                                                                                                                                          | Examples of ownership                                                                      |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Structural design / computational mechanics</b>        | Conceptual and preliminary design; analysis models; solver integration; code checks; option comparison; structural optimisation under engineering constraints. | New-design benchmark; WG task; design and verification module.                             |
| <b>Existing structures / reliability / retrofit</b>       | Condition and safety assessment; residual capacity; uncertainty; reliability; repair, strengthening and retrofit decision-making.                              | Existing-structure benchmark; assessment task; retrofit and risk module.                   |
| <b>AI, LLMs and computational software</b>                | Agent architectures; tool use; evaluation; guardrails; trusted solver interfaces; reproducibility.                                                             | Agent evaluation task; tool-integration benchmark; failure-testing exercise.               |
| <b>Structural identification / SHM / digital twins</b>    | System identification; model updating; monitoring-to-assessment; provenance; lifecycle decisions; as-designed and as-observed information.                     | Existing-system benchmark; digital-twin task; identification and data-provenance guidance. |
| <b>Autonomous inspection / robotics</b>                   | UAVs and mobile robots; NDT integration; adaptive inspection planning; human-robot interaction; selected construction and as-built verification tasks.         | Cyber-physical benchmark; inspection case; robotics training module.                       |
| <b>Human factors, assurance and standards</b>             | Human performance baselines; trust calibration; review and override; decision authority; governance; professional accountability; acceptance evidence.         | Human–AI reliability study; oversight framework; assurance case; standards roadmap.        |
| <b>Owners, public authorities, consultancies and SMEs</b> | Real workflows; procurement; asset priorities; validation needs; adoption constraints.                                                                         | Practice challenge; stakeholder workshop; policy or practice brief.                        |

### Benefits of participation

Participation offers access to an interdisciplinary European network, opportunities to lead tasks and benchmarks, mobility and Training School activities, closer contact with infrastructure owners and professional practice, and a shared platform for future research, education and standardisation initiatives.

Joining at the proposer stage gives partners the opportunity to shape the objectives, Working Group tasks, benchmark programme and education plan before submission.

### Network balance

The 2026 call requires at least seven COST Full or Cooperating Members in the proposer network, with at least 50% of represented countries from Inclusiveness Target Countries and at least 40% Young Researchers and Innovators. Beyond these formal requirements, RELIANCE will seek scientific, geographical, career-stage and sectoral balance, with each proposer linked to a concrete scientific, educational or stakeholder contribution.



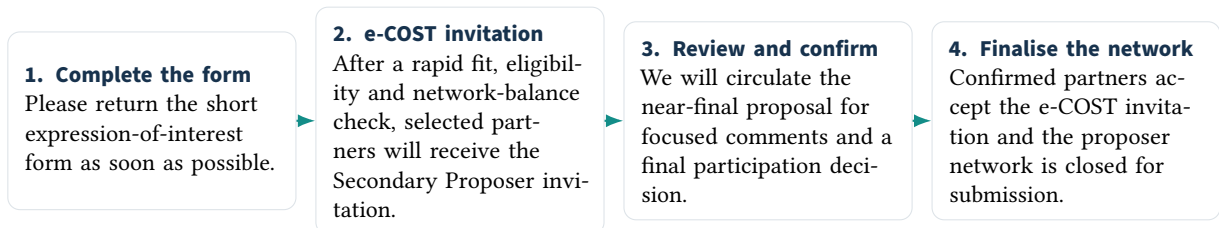
## 5. Joining the proposal

**Interested in joining RELIANCE? Please complete the short expression-of-interest form sent with this concept note.** The form gives us the information needed to assemble a balanced proposer network and identify where each partner can contribute most effectively.

### What we ask in the form

|                                      |                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Name, affiliation and country</b> | Basic information for the proposer network.                                                                                   |
| <b>Relevant expertise</b>            | A short description of the methods, applications or perspective you would bring.                                              |
| <b>Preferred contribution</b>        | Preferred WG or task, benchmark problem, stakeholder activity or Training School contribution.                                |
| <b>COST profile</b>                  | YRI status and whether you are already Main or Secondary Proposer in another 2026 COST proposal.                              |
| <b>Stakeholder connections</b>       | Relevant owners, consultancies, SMEs, public authorities, standards bodies, software providers or professional organisations. |
| <b>Leadership interest</b>           | Optional interest in task coordination, education, YRI and ITC activities, stakeholder liaison or communication.              |

### Next steps



**Timing and decision point.** Expressions of interest will be reviewed on a rolling basis. Because the timetable is short, we will use the form to assess quickly whether the network has the scientific critical mass and balance needed to proceed. We aim to take an early go/no-go decision rather than keep recruitment open until the submission deadline. Once a viable network is assembled, further invitations will be limited to specific expertise or balance gaps.

**Key 2026 dates and rules.** Submission closes on **28 October 2026 at 12:00 CET**. The proposer network must include at least seven COST Full or Cooperating Members; at least 50% of represented countries must be ITCs; at least 40% of individual proposers must be YRIs; and an individual may act as Main or Secondary Proposer in only one proposal in the 2026 Open Call. Under the COST 2026 rules, a Secondary Proposer cannot withdraw after accepting an invitation in the same Collection; acceptance should therefore follow the final review and confirmation step.

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### References

- COST Association, *Open Call for Proposals - Collection OC-2026-1 and Open Call Proposers' Guidelines OC-2026-1*. [cost.eu/funding/documents-guidelines](https://cost.eu/funding/documents-guidelines)
- COST Action CA25158, *Tech-TO.BE*; COST Action CA25178, *API-Net*. [cost.eu/actions/CA25178](https://cost.eu/actions/CA25178)
- Chen, J. and Bao, Y. (2025), multi-agent LLM framework for code-compliant reinforced-concrete design, *Automation in Construction*, 177, 106331.
- Sharma, A. and Bhowmik, B. (2026), agentic AI for physics-informed structural health monitoring, *Advanced Engineering Informatics*, 70, 104224.
- Panigati, T. et al. (2025), "Drone-based bridge inspections: Current practices and future directions", *Automation in Construction*, 173, 106101.
- Li, X. et al. (2026), closed-loop robotic inspection and digital twins for fatigue prognosis of in-service steel bridges, *Communications Engineering*, 5, 76.
- Wu, L. et al. (2026), autonomous structural inspection and damage assessment using agentic AI and aerial robotics, *AI in Civil Engineering*, 5, 23.
- Slezák, S. et al. (2026), multi-agent LLM architecture for human-robot collaboration in building disassembly, *Construction Robotics*, 10, 23.
- Regulation (EU) 2024/1689; ISO/IEC 23894:2023; ISO/IEC 42001:2023.